

Roots-of-unity lower bounds for finite-dimensional complex Littlewood constants

A partial result for arXiv:1612.08397

Automated mathematics run `fa_banach_001`, lane 13

9 August 2026

Status. Candidate partial result; current verdict: *likely valid*. Human review is recommended. For every $n \geq 2$ this packet gives an explicit lower bound for both finite-dimensional complex constants in the source problem and proves that their common value 1 at $n = 2$ cannot persist for any $n \geq 3$. It does not determine the exact constants beyond dimension two.

1 Source question

The source is W. Cavalcante, D. Pellegrino, and E. Teixeira, *On the geometry of multilinear forms*, arXiv:1612.08397, Section 5.2, page 17 [1]. For a bilinear form $T : \ell_\infty^n(\mathbb{C}) \times \ell_\infty^n(\mathbb{C}) \rightarrow \mathbb{C}$, write $a_{ij} = T(e_i, e_j)$ and

$$\|T\| = \max_{|x_j| \leq 1, |y_i| \leq 1} \left| \sum_{i,j=1}^n a_{ij} y_i x_j \right| = \max_{|x_j| \leq 1} \sum_{i=1}^n \left| \sum_{j=1}^n a_{ij} x_j \right|.$$

The optimal constants are defined by

$$\begin{aligned} \left(\sum_{i=1}^n \left(\sum_{j=1}^n |a_{ij}| \right)^2 \right)^{1/2} &\leq \kappa_O^{\mathbb{C}}(n) \|T\|, \\ \sum_{i=1}^n \left(\sum_{j=1}^n |a_{ij}|^2 \right)^{1/2} &\leq \kappa_L^{\mathbb{C}}(n) \|T\|. \end{aligned}$$

The source identifies their exact finite-dimensional values as an open problem and proves in Theorem 25 that

$$\kappa_O^{\mathbb{C}}(2) = \kappa_L^{\mathbb{C}}(2) = 1.$$

2 Proof intuition

The most economical way to force a large mixed numerator is to give every row two unimodular entries. Arrange their relative phases uniformly around the circle: row k is $(1, \zeta^k)$ for an n th root of unity ζ . The mixed numerator is then $n\sqrt{2}$, while the bilinear norm becomes the maximum of a single equally spaced absolute-cosine sum.

For each positive integer n , following the notation used by [1], let $\kappa_O^{\mathbb{K}}(n)$, $\kappa_L^{\mathbb{K}}(n)$, $\kappa_{4/3}^{\mathbb{K}}(n)$ be extrema constants for the following inequalities:

$$\left(\sum_{i=1}^n \left(\sum_{j=1}^n |T(e_i, e_j)| \right)^2 \right)^{1/2} \leq \kappa_O^{\mathbb{K}}(n) \|T\|,$$

$$\sum_{i=1}^n \left(\sum_{j=1}^n |T(e_i, e_j)|^2 \right)^{1/2} \leq \kappa_L^{\mathbb{K}}(n) \|T\|,$$

and

$$\left(\sum_{i,j=1}^n |T(e_i, e_j)|^{4/3} \right)^{3/4} \leq \kappa_{4/3}^{\mathbb{K}}(n) \|T\|$$

for all bilinear forms $T: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$. Classical inequalities, see [1] [15] [17], due to Orlicz and Littlewood assert that

$$\begin{aligned} \kappa_O^{\mathbb{C}}(\infty) &:= \lim_{n \rightarrow \infty} \kappa_O^{\mathbb{C}}(n) < \infty, \\ \kappa_L^{\mathbb{C}}(\infty) &:= \lim_{n \rightarrow \infty} \kappa_L^{\mathbb{C}}(n) < \infty, \\ \kappa_{4/3}^{\mathbb{C}}(\infty) &:= \lim_{n \rightarrow \infty} \kappa_{4/3}^{\mathbb{C}}(n) < \infty. \end{aligned}$$

The exact values of $\kappa_O^{\mathbb{C}}(n)$ and $\kappa_L^{\mathbb{C}}(n)$ are stated as an open problem in [1] Page 31]. We solve this problem here for $n = 2$, with the aid of techniques introduced by Jameson, [12], concerning unital bilinear forms when dealing with a specific form of two-dimensional Grothendieck's inequality. We will ultimately prove:

Theorem 25. $\kappa_O^{\mathbb{C}}(2) = \kappa_L^{\mathbb{C}}(2) = \kappa_{4/3}^{\mathbb{C}}(2) = 1$.

Proof. Let A, B be complex C^* -algebras with identities e_A, e_B . According to [12] we say that a bilinear form $V: A \times B \rightarrow \mathbb{C}$ is unital if

Figure 1: Definitions, open-problem statement, and the dimension-two theorem on page 17 of the source paper.

That trigonometric maximum is exactly computable. On half a fundamental period the signs of the cosines change only once. Removing the absolute values therefore turns the sum into a signed geometric series. Its value is real when n is odd and has a simple phase shift when n is even; both cases have amplitude $\csc(\pi/(2n))$. Transposing the matrix preserves the bilinear norm and converts the same example into one with equally large Orlicz numerator.

3 Result

Theorem 1 (Explicit roots-of-unity lower bounds). *For every integer $n \geq 2$,*

$$\kappa_O^{\mathbb{C}}(n) \geq \frac{n}{\sqrt{2}} \sin\left(\frac{\pi}{2n}\right), \quad \kappa_L^{\mathbb{C}}(n) \geq \frac{n}{\sqrt{2}} \sin\left(\frac{\pi}{2n}\right).$$

Consequently, for every $n \geq 3$ both constants are strictly greater than 1. In particular,

$$\kappa_O^C(3), \kappa_L^C(3) \geq \frac{3\sqrt{2}}{4}.$$

The displayed lower bound increases with n and converges to $\pi/(2\sqrt{2})$.

The proof uses the following exact elementary identity.

Lemma 1 (Equally spaced absolute-cosine sum). *For every integer $n \geq 2$,*

$$\max_{u \in \mathbb{R}} \sum_{k=0}^{n-1} \left| \cos\left(u + \frac{\pi k}{n}\right) \right| = \csc\left(\frac{\pi}{2n}\right).$$

If n is odd, a maximum occurs at $u = 0$; if n is even, one occurs at $u = \pi/(2n)$.

Proof. Put $\alpha = \pi/n$ and

$$S_n(u) = \sum_{k=0}^{n-1} |\cos(u + k\alpha)|.$$

Reindexing the sum shows that S_n is α -periodic and even. It is therefore enough to take $0 \leq u \leq \alpha/2$.

First let $n = 2m + 1$. On this interval the cosines have positive sign for $0 \leq k \leq m$ and negative sign for $m + 1 \leq k \leq 2m$ (with a harmless zero at an endpoint). The geometric-sum identity gives

$$\sum_{k=0}^m e^{ik\alpha} - \sum_{k=m+1}^{2m} e^{ik\alpha} = \csc(\alpha/2).$$

Taking the real part after multiplication by e^{iu} yields

$$S_n(u) = \csc(\alpha/2) \cos u.$$

Its maximum on the chosen interval occurs at $u = 0$.

Now let $n = 2m$. The signs are positive for $0 \leq k \leq m - 1$ and negative for $m \leq k \leq 2m - 1$. This time

$$\sum_{k=0}^{m-1} e^{ik\alpha} - \sum_{k=m}^{2m-1} e^{ik\alpha} = \cot(\alpha/2) - i.$$

Hence

$$\begin{aligned} S_n(u) &= \cot(\alpha/2) \cos u + \sin u \\ &= \csc(\alpha/2) \cos(u - \alpha/2), \end{aligned}$$

whose maximum occurs at $u = \alpha/2$. Both cases prove the identity. $\square$

Proof of Theorem 1. Let $\zeta = e^{2\pi i/n}$ and let $A_n = (a_{kj})$ be the $n \times n$ matrix whose only nonzero columns are

$$a_{k1} = 1, \quad a_{k2} = \zeta^k, \quad 0 \leq k \leq n-1.$$

The second formula for the bilinear norm above gives

$$\|A_n\| = \max_{|x_1|, |x_2| \leq 1} \sum_{k=0}^{n-1} |x_1 + \zeta^k x_2|.$$

The objective is separately convex in x_1 and x_2 , with each disk the convex hull of its unit circle. Thus a maximum can be chosen with $|x_1| = |x_2| = 1$. After removing a common phase and writing $x_2/x_1 = e^{it}$, we obtain

$$\begin{aligned}\|A_n\| &= \max_{t \in \mathbb{R}} \sum_{k=0}^{n-1} |1 + e^{i(t+2\pi k/n)}| \\ &= 2 \max_{u \in \mathbb{R}} \sum_{k=0}^{n-1} \left| \cos\left(u + \frac{\pi k}{n}\right) \right| \\ &= 2 \csc\left(\frac{\pi}{2n}\right),\end{aligned}$$

where the last equality is Lemma 1.

Every row of A_n has Euclidean norm $\sqrt{2}$. Therefore its mixed Littlewood numerator is $n\sqrt{2}$, and

$$\kappa_L^{\mathbb{C}}(n) \geq \frac{n\sqrt{2}}{2 \csc(\pi/(2n))} = \frac{n}{\sqrt{2}} \sin\left(\frac{\pi}{2n}\right).$$

For the Orlicz inequality, use A_n^T . Transposition preserves the bilinear norm. The first two rows of A_n^T have ℓ_1 norm n , and all remaining rows vanish, so its Orlicz numerator is $\sqrt{n^2 + n^2} = n\sqrt{2}$. This proves the second lower bound.

At $n = 2$ the bound is 1. To see strict growth thereafter, set $f(x) = x \sin(\pi/(2x))$ and $t = \pi/(2x)$. Then

$$f'(x) = \sin t - t \cos t > 0,$$

because $\sin t - t \cos t$ vanishes at $t = 0$ and has derivative $t \sin t > 0$. Finally,

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{2}} \sin\left(\frac{\pi}{2n}\right) = \frac{\pi}{2\sqrt{2}}.$$

□

4 Verification, limitations, and novelty

The script `code/verify_roots_of_unity_bound.py` samples a fundamental half-period of S_n , checks the exact norm formula, and compares the resulting matrix ratio with the theorem for $2 \leq n \leq 12$. This is only numerical corroboration; the proof above is independent of it.

The theorem does not determine either exact finite-dimensional constant for $n \geq 3$. The first unresolved candidate equality is

$$\kappa_O^{\mathbb{C}}(3) = \kappa_L^{\mathbb{C}}(3) = \frac{3\sqrt{2}}{4}.$$

A bounded attempt to prove the necessary upper bound found no valid extension of the source's dimension-two classification. The exact missing inequality is documented in the accompanying dimension-three frontier attempt note.

A later paper of Cavalcante, Núñez-Alarcón, Pellegrino, and Rueda proves the dimension-free complex mixed constant $2/\sqrt{\pi}$ [2]. Thus the present asymptotic lower bound $\pi/(2\sqrt{2})$ lies below the known dimension-free optimum, as it must. That later result does not provide the finite- n formula requested by the source.

The bounded novelty audit on 9 August 2026 searched the arXiv full text and citation neighborhood of arXiv:1612.08397, exact variants of the displayed formula, and combinations of “roots of

unity,” “complex Littlewood,” and “finite-dimensional constants.” It also checked arXiv:1912.10313. No match for this construction or all- n lower bound was located. Novelty is therefore provisional, not certified.

Human-review recommendation. The result is short enough for direct verification. Review should focus on the two signed geometric-sum identities in Lemma 1 and on the separate-convexity reduction from the bidisk to the torus.

References

- [1] W. Cavalcante, D. Pellegrino, and E. Teixeira, *On the geometry of multilinear forms*, arXiv:1612.08397 (2016), Section 5.2.
- [2] W. Cavalcante, D. Núñez-Alarcón, D. Pellegrino, and P. Rueda, *Optimal constants of classical inequalities for complex scalars*, arXiv:1912.10313 (2019).